

Plants utilize nitrate as their source of nitrogen. Therefore, microorganisms play essential roles in providing nitrogen in bioavailable forms for plants and animals.

- **Denitrification**

In the absence of atmospheric oxygen, some microorganism use nitrate and reduces it into N_2 . This process is called denitrification. Denitrification causes loss of nitrogen into the atmosphere. Therefore, it reduces available nitrogen in the soil. Denitrification frequently occurs in waterlogged soils, where limited oxygen is present. *Pseudomonas* sp converts nitrate ions into molecular nitrogen through following steps; nitrate to nitrite, nitrite to nitrous oxide and nitrous oxide to nitrogen gas



- **Nitrogen fixation**

The process of conversion of nitrogen gas to ammonia is termed as nitrogen fixation. Bacteria that are able to fix nitrogen require nitrogenase enzyme. Nitrogenase enzyme is inactivated by oxygen. There are two types of nitrogen fixing microorganisms, free-living and symbiotic.

- o **Free-living nitrogen fixing bacteria-** They are abundantly found in the rhizosphere. Rhizosphere is the soil immediately surrounding plant root. e.g. *Azotobacter* sp. Many cyanobacteria are also able to fix nitrogen. e.g. *Nostoc*. These organisms, exhibit mechanisms to prevent the exposure of nitrogenase enzyme to the atmospheric oxygen; *Azotobacter* sp - high rate of aerobic respiration, Cyanobacteria- heterocyst. Some anaerobic bacteria such as *Clostridium* sp also fixes nitrogen.
- o **Symbiotic nitrogen-fixing microorganism-** They play an important role in agricultural crops such as leguminous crops; soybean, beans, peas and peanuts. These symbiotic nitrogen-fixing bacteria are commonly known as rhizobia. Leguminous plants are specially adapted to facilitate symbiotic nitrogen fixation. Plants form root nodules where, anaerobic conditions and nutrients for bacteria are given. Bacteria fixes nitrogen and make it bio available to be used by the plant. Some non-leguminous plants can also fix nitrogen symbiotically with different microbial combinations. Lichen, a combination of fungus and an algae or cyanobacteria also fixes nitrogen. In rice paddy waters a small free-floating fern, *Azolla* fixes nitrogen symbiotically with *Anabaena* sp.

Interactions of soil microorganisms relevant to plant growth

Microorganisms in the soil directly interact with plants. Interactions in the soil such as rhizosphere, mycorrhizae and endophytes with symbiotic relationship of bacteria, fungi and plant cells. These microorganisms are beneficial to the plants in numerous ways such as nitrogen fixation, increased uptake of water and nutrients such as phosphorous, secretion of plant hormones such as indole acetic acid, increases iron uptake at iron limiting conditions and defense against pathogens. In return plants provide organic compounds essential for the microorganisms.

Rhizosphere

This is a kind of symbiotic interaction between plant roots, and soil surrounding the root surface for about few millimeters. This micro-ecological zone is called rhizosphere. Rhizosphere is considered as the most biodiverse and dynamic habitat on earth.

Microorganisms in the rhizosphere feed on the compounds exuded by roots such as sugars, amino acids and various aromatic compounds. Microorganisms compete –antagonize using antimicrobial compounds with each other for resources such as nutrients, space, and water in the rhizosphere. Bacteria are the most numerous organisms in the rhizosphere. Three most common genera of bacteria inhabiting rhizosphere are *Pseudomonas*, *Bacillus* and *Agrobacterium*. Root exudates act as chemical signals for bacteria to move towards the root surface. Both pathogenic and symbiotic fungi associate with the rhizosphere.

Mycorrhizae

Mycorrhizae (myco = fungus, rhiza = root) are symbiotic association between plant roots with fungi. All most all land plants have symbiotic association with one or more mycorrhizal fungi. Mycorrhizal fungi extend the surface area over which nutrients and water can be taken up by the plant. They can reach small pores in soil where plant roots cannot reach and uptake nutrients. Most significantly, mycorrhizae increase uptake of immobile nutrients such as phosphorous, zinc and copper. In return, mycorrhiza receive organic carbon from the plant.

Role of soil microorganisms in improving soil quality

Free living soil microorganisms and those associated with root surfaces, play a major role in improving soil quality. Microorganisms are involved in the formation of stable soil-aggregates, which are characteristics of good soil structure in fertile soils. Fungal filaments, Actinomycetes filaments and polysaccharide gums/slimes produced by bacteria are involved in soil aggregation formation.

Microbiology of domestic water and waste water

Contamination routes of drinking water

Drinking water resources can be contaminated with infectious disease causing organisms or chemical pollutants. Since water undergoes a filtering of microorganisms when pass through the soil into deeper layers, generally, water from springs and deep wells are of good quality in microbial point of view. Drinking water can be contaminated by dangerous pathogens when feces enter the water supply. Many diseases are disseminated by the fecal - oral route of transmission, in which, pathogens are shed in human or animal feces, contaminate water, and ingested via drinking. Examples for diseases spread by water (water-borne diseases) are typhoid fever, cholera, diarrhea etc.

Contamination of drinking water by chemical pollutants is a global issue. Industry, domestic and agriculture sectors release a large amount of chemicals that leach from the soil surface to groundwater. Most of these chemicals are resistant to biodegradation. Most freshwaters, such as tanks, often contain excessive amounts of nitrates and phosphates that come from agricultural fertilizer and household chemicals such as detergents. Accumulation of excessive amounts of such nutrients cause eutrophication, and flourish the growth of cyanobacteria and algae. They are toxic to humans. This kind of dense growth of cyanobacteria and algae, is called algal blooms. Various industries also release chemicals that are non-biodegradable. They too can contaminate drinking water supplies.

Microorganisms as indicators of water quality

Water supplies can be contaminated with pathogenic microorganisms such as *Salmonella* spp., *Shigella* sp. and *Vibrio* sp. that causes infectious diseases such as typhoid and Cholera. Therefore, it is essential to detect the presence of such microorganisms before the consumption to prevent disease outbreak.

However, it is not practical to test water samples to look for pathogens because, pathogens may present only in small numbers and might not be included in the test sample. On the other hand, since the testing for pathogens take time, discovery of pathogens in the laboratory takes time and, it will be too late to prevent an outbreak of the disease. Therefore, it is important to test water samples routinely for "indicator organisms, which will indicate potential contamination of water supply by pathogens.

One of the major criteria for an indicator organism is that it should be consistently present in human feces in large numbers. This ensures the, presence of indicator organism providing evidence for contamination of water supply with human feces.

Sri Lanka, and many other countries use coliform bacteria as an indicator organism to test drinking water quality. Coliforms are aerobic or facultative anaerobic, gram-negative, non-endospore forming, rod shaped bacteria that ferment lactose to form gas within 48 hours in a lactose broth medium at 35 °C. Coliforms constitute a large portion of human intestinal microflora. They are nonpathogenic line in the intestine. Therefore, presence of coliforms in water indicates fecal contamination. However, there are some coliforms found in plant and soil samples. There are specialized test to distinguish coliforms that has fecal origin from coliforms present in plant and soil samples. Coliform test is routinely carried out by laboratories of the national water supply and drainage board to determine water quality of drinking water.

Water-borne diseases

Pathogenic organisms that are frequently transmitted through water cause infections in the intestinal tract such as typhoid, paratyphoid, cholera, gastroenteritis and dysentery.

Process of drinking water treatment

Drinking water that comes from various water supplies can be polluted at any time. Therefore, water needs to be purified before consumption for our health and safety. The idea of water purification is not to sterilize water, but to make water free of disease-causing organisms. There are three main steps in an urban water treatment plant. They are,

- Sedimentation and coagulation,
- Filtration
- Disinfection.

I. Sedimentation & Coagulation

This is the first step. If water is turbid (cloudy), water is allowed to stand in a holding tank for a period of time to allow settling down of particles suspended in water. This occurs in large reservoirs, where water remains for a holding period during which large particulate matter settles to the bottom. Sedimentation is enhanced by adding alum (Aluminum potassium sulphate) which produces a sticky precipitate. Many microorganisms as well as finely

suspended matter are removed in this manner.

II. Filtration

After sedimentation and coagulation, water is filtered by passing through beds of fine sand. Filtration removes protozoan cysts and other microorganism. Microorganisms are trapped by surface absorption onto the sand particles. This removes about 99% of bacteria. Some urban treatment plants additionally use activated carbon for the removal of toxic chemicals.

III. Disinfection

The final step in water treatment is disinfection. Water can be disinfected by several methods. Chlorination is one of the most commonly used method. It kills pathogenic bacteria. Disinfection by ozone (O_3) is another method. Ozone is highly reactive. It kills microorganisms by oxidation. Ozone has become a preferred method of disinfection since it leaves no taste or odor and has little residual effect.

Wastewater (sewage) management

Wastewater includes water from household use such as water used for toilets and washing, water from urban drainage systems and from industry. Many developing countries and some developed counties still do not have proper wastewater treatment mechanisms.

• Principles and main steps in purification of industrial waste water

I. Primary treatment- Following steps are involved in the primary treatment.

- Large floating materials are screened out
- Removal of sand
- Removal of oil and grease
- Solid matter settles out in sedimentation tanks
- Sludge collected and removed
- No biological activity is used here
- Primary treatment removes 25-35% organic matter

II. Secondary treatment- Following steps are involved in the Secondary treatment.

- The liquid flowing out of primary treatment, enters the secondary treatment stage.
- During this treatment, waste water is aerated to facilitate the growth of aerobic bacteria and rapid microbial oxidation. One of two systems used are, activated sludge and trickling filter.
- In activated sludge, vigorous aeration is done mechanically. In trickling filters, waste water is slowly sprinkled or sprayed over a bed of rocky material and allowed to trickle. In this process, microorganisms grow on the filter bed and oxidize organic matter.
- In secondary treatment 75-95 % of organic material is oxidized.
- The liquid flowing through these systems are then disinfected and allowed to flow

into natural waters.

- Sludge remaining from both treatments is sent to an anaerobic sludge digester where anaerobic decomposition convert the organic material in the sludge, finally to methane and CO₂ (biogas).
- Digested sludge can be used as fertilizer.

Adverse effects of discharging large amount of waste water into natural water bodies.

- Dissemination of pathogenic microorganisms
- Water pollution due to accumulation of biodegradable material and their decomposition products
- Decomposition which may consume large amounts of oxygen in water affecting aquatic organisms (high BOD – biological oxygen demand)
- Anaerobic decomposition leading to bad smells

Solid waste treatments

Nature of solid wastes

Solid waste include materials such as plant and animal residues, food waste, paper, plastic, polythene and glass. Among these, organic wastes such as plant and animal residues and food wastes are rapidly degradable. However, synthetic materials such as plastic and polythene are not easily degraded and tend to accumulate continuously. Proper management of solid waste, ensure community health and environmental safety. Accumulation of unmanaged waste in big piles in the open environment can lead to soil, air and water pollution which is harmful for the environment and organisms in contact with the polluted environment.

Environmental and hygienic importance of recycling solid wastes

- Open dumping of waste provide breeding grounds for mosquitoes, flies, other insects and rats. These organisms serve as vectors for dangerous diseases such as dengue, chikungunya, various food borne diseases and leptospirosis.
- Contaminated water resources have the risk of spreading water borne diseases such as typhoid, paratyphoid, cholera, dysentery and gastroenteritis.
- Piles of waste at public and residential area often create social issues due to unpleasant smell generated during anaerobic digestion of waste.
- Heavy piles of waste sometimes can be dangerous due to accumulation of methane, generated during anaerobic digestion of waste. Accumulation of methane can cause explosion and fire.
- Groundwater can be polluted by the leachate of waste piles. Leachate means liquid pass through the waste, extracting the dissolved and the suspended matter from the waste.

Therefore, solid waste must be properly managed using appropriate technology which is socially accepted and environment friendly.

Methods of minimizing problems created by solid wastes (solid waste management)**1. Sorting and recycling**

In many countries municipal solid waste is sorted into kitchen waste, plant material, paper, plastic, glass etc. and collected into separate containers. Sorting allows separation of waste that can be recycled such as paper, plastic and glass. Although paper is presumably biodegradable, it is not easily degraded since microorganisms cannot effectively attack papers when the waste is placed into large compacted dumping sites.

2. Degradation/ decomposition of organic matter

Readily degradable organic matter such as kitchen and garden waste is subjected to natural microbial degradation by the process of composting. Resulting compost is used in agriculture and gardening. Compost generate additional income to the waste management authorities such as municipal council. Anaerobic condition in the pile of compost, promote the activity of methanogenic bacteria. The methane gas they produce can be collected to generate electricity.

3. Sanitary land filling

Sanitary landfills are one of the most popular forms of waste disposal, primarily because, they are the least expensive way to dispose the waste. More than four-fifths of municipal solid waste is disposed in landfills. A sanitary landfill is an engineered means of disposal of waste. In a sanitary landfill, waste is spread in layers on a piece of property, usually on marginal or sub marginal land. The objective is to spread the layers and then compact them tightly, greatly reducing the volume of the waste. The waste is then covered by soil. A landfill should not be located in areas with high ground water levels. Much of the waste in a sanitary landfill will decompose through biological and chemical processes that produce solid, liquid and gaseous products.

Microorganisms and food**Why food is spoiled by microorganisms**

All food materials used for human consumption are of plant or animal origin. Microorganisms inhabit nearly every niche on earth, and food supply which are of plant or animal origin is no exception. Food have a natural micro biota and also get contaminated with microorganisms in nature during handling and processing. Food materials contain nutrients and water, act as culture media for growth of microorganisms. The nutrients present in the food materials, the same that is required for the growth of many microorganisms. Therefore, food serves as a culture medium such as nutrient broth. Numerous bacteria, yeast and , molds will grow in this broth.

Physical, chemical and biological changes occur in food spoilage

Growth of microorganisms in food changes physical, chemical and biological structures of food and making them unfit for consumption which is termed as food spoilage. Microorganisms growing in food are heterotrophic bacteria and fungi. During this process, they breakdown carbohydrates, proteins and fats, to obtain energy and other requirements for their own growth. Various extra cellular enzymes such as amylase pectinase, cellulase, protease and lipase are secreted by contaminant microorganisms, in order to complete the above process. As a result of this, major constituents of food, undergo chemical, physical and biological changes.

Chemical changes take place in food**1. Putrefaction**

Protein present in the food source is broken down by the proteolytic enzymes secreted by proteolytic microorganisms into Amino acids, Amines, Ammonia and hydrogen sulphide (H₂S).

2. Fermentation

Complex carbohydrates present in the food source are broken down into simple carbohydrates by amylase secreted from microorganisms. Simple carbohydrates (sugars) are then converted into carbohydrate food acid, alcohol and gases by enzymes secreted from saccharolytic microorganisms.

3. Rancidity

Lipids present in food source are converted into fatty acids and glycerol by enzymes secreted by lipolytic microorganisms.

Physical changes that may take place in food

1. Softening of food
2. Pigmentation
3. Ropiness
4. Slime or gum formation (polysaccharide)
5. Toxin accumulation

External factors influence on food spoilage

External factors are environmental factors that affect both food and microorganisms.

1. Temperature of storage- Growth of microorganisms is affected by a wide range of temperatures, at low temperature growth is slow, therefore, spoilage is also slow, at ambient temperature growth is high, and therefore, spoilage is also high. However, even at low temperature (e.g. food stored at 4 °C in the refrigerator getting spoiled) microbial spoilage occurs by psychrophilic bacteria.
2. Relative humidity (RH) of environment- Relative humidity of the storage environment is important, as it relates to the availability of moisture, which is an important factor for food spoilage due to microbial growth. Low moisture containing food should not be stored in high RH environments, as the food will pick moisture from the atmosphere and the microbial growth will be facilitated.
3. Presence and concentrations of gases in the environment- presence or absence of O₂ determine the type of contaminants causing food spoilage. There are both aerobic as well as anaerobic microbes cause food spoilage.

Internal factors influence on food spoilage

Internal factors are those factors that are present in food itself.

1. pH- Most microorganisms grow best at pH values around 7.0 (6.6 – 7.5) and very few grow below 4.0. In general, molds and yeasts can grow over a wide range of pH from very low to very high (pH 2 - 10). Bacteria in general, grow between pH 5 – 7. Fruits such as limes, orange and banana are likely to be spoiled by molds and yeast. Most animal foods such as beef, chicken, fish, milk are susceptible to bacterial, mold and yeast spoilage.
2. moisture content- Drying is the oldest method of preserving food which is based on reducing moisture content. High moisture containing food such as meat and fish are spoiled by bacteria. Low moisture containing food such as biscuits and bread are spoiled by molds. Dried milk powder or flour which contains very low moisture are not easily spoiled by bacteria or molds. Salt and sugar containing foods (available water is low) are generally spoiled by halophilic bacteria (salted foods), osmophilic and xerophilic molds/ yeast (sugary foods).
3. Nutrient content- Water, source of energy, nitrogen, vitamins and minerals are essential nutrients for growth of microorganisms. Nutrient rich food are easily spoiled by microorganisms. e.g.; Milk, meat
4. Biological structure- The natural covering of some foods protect the food from the entry and the damage cause by the spoilage microorganisms. e.g., outer covering of fruits, shell of egg

Food borne pathogens of humans

- Typhoid - *Salmonella typhi*
- Dysentery - *Shigella*
- Cholera - *Vibrio cholera*
- Food poisoning - *Staphylococcus aureus*
- Botulism - *Clostridium botulinum*
- Aflatoxin - *Aspergillus flavus*

Effects of food spoilage on human health

Some microorganisms produce various toxic substances in food, which cause food poisoning or food intoxication when consumed. Illness may result from consuming food spoiled by microorganisms. Microorganisms grow and multiply in food, increasing the numbers of microbial cells, and also produce toxic chemical substances. Ingestion of large numbers of microbial cells and their toxic chemicals by any individual through highly contaminated food, may develop disease. These diseases/ illnesses are of two kinds;

- (a) food borne infections- In food borne infections, the contaminating pathogenic microorganisms enter the body of the individual consumed spoiled food, and inside the host, they grow, multiply and also produce toxins which cause symptoms of the characteristic disease.

e.g. *Salmonella typhi* - typhoid fever

Shigella - dysentery

Vibrio cholerae - cholera

(b) Food intoxication- In food intoxication, the spoiled food contains toxins, produced due to microbial growth. Any individual who consumes such contaminated food containing toxins will develop symptoms of the disease, within a short period of time. e.g;

- Food poisoning - *Staphylococcus aureus*
- Botulism - *Clostridium botulinum*
- Aflatoxins - Produced by fungi *Aspergillus flavus*

In general all the above illnesses are referred to as due to food poisoning.

(c) Viruses also (entero viruses) cause some food borne illnesses.

References

- Tortora, G.J., Funke, B. R., & Case, C. L. (2011). *Microbiology, An Introduction*. Pearson Education Ltd.

Notes:

This is to acknowledge that some of the diagrams used in this book have been taken from various electronic sources using internet . This book is not published to make profit and sold only to cover cost.

The resource book is prepared according to the subject content and learning outcomes of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.

The content of this Resource book declares the limitation of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.